COMP 451
Introduction to Natural Language Processing
Fall 2024

Project#1

Due Date: December 9, 2024 – 23:59 pm

I) Description:

In this project, you are expected to explore the use of prompt engineering for data annotation, focusing on four NLP tasks: i) Named Entity Recognition (NER), ii) Aspect-Based Sentiment Analysis (ABSA), iii) Coreference Resolution (COREF), and iv) Question Answering (Q/A). For the task that you select, follow the steps given below:

i) Find an annotated dataset from the web *(Dataset1)*

ii) Take 100-200 samples from the dataset (Dataset1), obtain its raw version and annotate it by using zero-shot, few-shot, and chain-of-thought prompting, Use ChatGPT, Claude, and Gemini. Prompts should be written in English.

iii) Compare the best set of annotations that you obtain with the original annotations and report your similarity/success rate.

iv) Collect raw data from the web (web scraping) and annotate them by using the same prompts and the language model *(Dataset2).*

v) Format Dataset 2 in JSON format.

- <u>The classes for NER:</u> Person, Location, Organization, Time, Currency
- <u>The number of aspects for ABSA:</u> At least 4 aspects
- <u>The question types for Q/A:</u> Who, What, Where, When
- You should collect <u>at least 300 samples</u> and annotate them. For evaluating the success of your prompts, you can use publicly available NLP libraries (e.g., NLTK and spaCy)
- Research and analyze existing studies in the literature that are related to your topic
- Work either on Turkish or English
- Form groups of 3-4 students

II) Submissions:

Submit a single zip file to the Blackboard System (before due date) that includes:

- **Web Scraping Code:** Provide the code used for crawling data
- **Raw and Annotated Dataset 2:** Provide your dataset in JSON format
- **Final Report:** Write a report (at most 10 pages long) that shows the final prompts that are used for annotation, the web sites used for data collection, the description of how the crawler can be run, the success/similarity rate of prompts on Dataset 1.

IV) Late Policy:
Up to two days; -30 points the first day and -50 points the second day.

V) Plagiarism:
If plagiarism is detected, you will not receive any grade from the project.

VI) Grading:
Dataset: & Web Scraping Code: 70 points
Report: 30 points